Exercise 14.1 SOLUTION

Q.1. Let $\underline{u} = 3\underline{i} + 2\underline{j} - 5\underline{k}$, $\underline{v} = \underline{i} - 5\underline{j} - \underline{k}$ and $\underline{w} = -4\underline{i} - \underline{j} + 7\underline{k}$. **Find the following:**

i. $\underline{u} + 2\underline{v} + \underline{w}$

Sol: By using the value of $\underline{u}, \underline{v}$ and $\underline{w}$

$$\underline{u} + 2\underline{v} + \underline{w} = 3\underline{i} + 2\underline{j} - 5\underline{k} + 2\left(\underline{i} - 5\underline{j} - \underline{k}\right) + \left(-4\underline{i} - \underline{j} + 7\underline{k}\right)$$

$$= 3\underline{i} + 2\underline{j} - 5\underline{k} + 2\underline{i} - 10\underline{j} - 2\underline{k} - 4\underline{i} - \underline{j} + 7\underline{k} = \underline{i} - 9\underline{j} + 0\underline{k} = \underline{i} - 9\underline{j}$$

ii. $\underline{v} - 3\underline{w}$

Sol: By using the value of $\underline{v}$ and $\underline{w}$

$$\underline{v} - 3\underline{w} = \underline{i} - 5\underline{j} - \underline{k} - 3\left(-4\underline{i} - \underline{j} + 7\underline{k}\right) = \underline{i} - 5\underline{j} - \underline{k} + 12\underline{i} + 3\underline{j} - 21\underline{k} = 13\underline{i} - 2\underline{j} - 22\underline{k}$$

iii. $|3\underline{v} + \underline{w}|$

Sol: $3\underline{v} + \underline{w} = 3\left(\underline{i} - 5\underline{j} - \underline{k}\right) + \left(-4\underline{i} - \underline{j} + 7\underline{k}\right) = 3\underline{i} - 15\underline{j} - 3\underline{k} - 4\underline{i} - \underline{j} + 7\underline{k} = -\underline{i} - 16\underline{j} + 4\underline{k}$

$$|3\underline{v} + \underline{w}| = \sqrt{(-1)^2 + (-16)^2 + (4)^2} = \sqrt{1 + 256 + 16} = \sqrt{273}$$

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Q.2. Find the magnitude of the vector $\underline{v}$ and write the direction cosines of $\underline{v}$.

i. $\underline{v} = 3\underline{i} - 2\underline{j} + 6\underline{k}$

نوٹ:
Direction cosine معلوم کرنے کے لیے $\underline{i}, \underline{j}$ اور $\underline{k}$ کے Co-efficients کو magnitude سے تقسیم کریں الگ الگ

Sol: $|\underline{v}| = \sqrt{(3)^2 + (-2)^2 + (6)^2}$

$|\underline{v}| = \sqrt{9+4+36} = \sqrt{49} = 7$

$\cos\alpha = \frac{a}{|\underline{v}|}$, $\cos\beta = \frac{b}{|\underline{v}|}$, $\cos\gamma = \frac{c}{|\underline{v}|}$

$\cos\alpha = \frac{3}{7}$, $\cos\beta = \frac{-2}{7}$, $\cos\gamma = \frac{6}{7}$

ii. $\underline{v} = -4\underline{i} + 4\underline{j} + 2\underline{k}$

Sol: $|\underline{v}| = \sqrt{(-4)^2 + (4)^2 + (2)^2} = \sqrt{16+16+4} = \sqrt{36} = 6$

$\cos\alpha = \frac{-4}{6} = \frac{-2}{3}$, $\cos\beta = \frac{4}{6} = \frac{2}{3}$, $\cos\gamma = \frac{2}{6} = \frac{1}{3}$

iii. $\underline{v} = -6\underline{i} + 8\underline{j}$

Sol: $|\underline{v}| = \sqrt{(-6)^2 + (8)^2}$

$|\underline{v}| = \sqrt{36+64} = \sqrt{100} = 10$

$\cos\alpha = \frac{-6}{10}$, $\cos\beta = \frac{8}{7}$

Q.3. Find t, so that $\left|2\underline{i} + (t-1)\underline{j} + t\underline{k}\right| = \sqrt{13}$

Sol: $\sqrt{(2)^2 + (t-1)^2 + (t)^2} = \sqrt{13}$

$\sqrt{4 + t^2 + 1 - 2t + t^2} = \sqrt{13}$

$\sqrt{2t^2 - 2t + 5} = \sqrt{13}$

Squaring both sides

$2t^2 - 2t + 5 = 13$

$2t^2 - 2t + 5 - 13 = 0$

$2t^2 - 2t - 8 = 0$

$t^2 - t - 4 = 0$

$a = 1,\ b = -1,\ c = -4$

By using Quadratic formula

$$t = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-4)}}{2(1)} \Rightarrow t = \frac{1 \pm \sqrt{1+16}}{2} = \frac{1 \pm \sqrt{17}}{2} \Rightarrow t = \frac{1 \pm \sqrt{17}}{2}$$

Q.4. Find a unit vector in the direction of $\underline{v} = -\underline{i} + 4\underline{j} - 8\underline{k}$

Sol: $|\underline{v}| = \sqrt{(-1)^2 + (4)^2 + (-8)^2}$

$|\underline{v}| = \sqrt{1+16+64} = \sqrt{81} = 9$

$\underline{v} = \frac{\underline{v}}{|\underline{v}|} = -\frac{1}{9}\underline{i} + \frac{4}{9}\underline{j} - \frac{8}{9}\underline{k}$

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Q.5. If $\underline{u} = 2\underline{i} + \underline{j} - 3\underline{k}$, $\underline{v} = -\underline{i} + 4\underline{j} + 2\underline{k}$ and $\underline{w} = 3\underline{i} - 2\underline{j} + \underline{k}$ Find a unit vector parallel to $4\underline{u} - 3\underline{v} + 2\underline{w}$.

Sol:

$$4\underline{u} - 3\underline{v} + 2\underline{w} = 4\left(2\underline{i} + \underline{j} - 3\underline{k}\right) - 3\left(-\underline{i} + 4\underline{j} + 2\underline{k}\right) + 2\left(3\underline{i} - 2\underline{j} + \underline{k}\right)$$

$$= 8\underline{i} + 4\underline{j} - 12\underline{k} + 3\underline{i} - 12\underline{j} - 6\underline{k} + 6\underline{i} - 4\underline{j} + 2\underline{k}$$

$$= 17\underline{i} - 12\underline{j} - 16\underline{k}$$

$$\left|4\underline{u} - 3\underline{v} + 2\underline{w}\right| = \sqrt{(17)^2 + (-12)^2 + (-16)^2} = \sqrt{289 + 144 + 256} = \sqrt{689}$$

$$\hat{a} = \frac{4\underline{u} - 3\underline{v} + 2\underline{w}}{\left|4\underline{u} - 3\underline{v} + 2\underline{w}\right|} = \frac{17}{\sqrt{689}}\underline{i} - \frac{12}{\sqrt{689}}\underline{j} - \frac{16}{\sqrt{689}}\underline{k}$$

Q.6. Find a vector whose

i. Magnitude is 5 and is parallel to $3\underline{i} + 4\underline{j} - \underline{k}$

ii. Magnitude is 7 and is parallel to $-\underline{i} + \underline{j} + \underline{k}$

i. $3\underline{i} + 4\underline{j} - \underline{k}$

Sol: $\underline{v} = 3\underline{i} + 4\underline{j} - \underline{k}$

$$\left|\underline{v}\right| = \sqrt{(3)^2 + (4)^2 + (-1)^2} = \sqrt{9 + 16 + 1} = \sqrt{26}$$

$$\hat{v} = \frac{\underline{v}}{\left|\underline{v}\right|} = \frac{1}{\sqrt{26}}\left(3\underline{i} + 4\underline{j} - \underline{k}\right)$$

Given Magnitude is 5

Required Vector $= \underline{a} = 5.\hat{v} = \frac{5}{\sqrt{26}}\left(3\underline{i} + 4\underline{j} - \underline{k}\right) = \frac{15}{\sqrt{26}}\underline{i} + \frac{20}{\sqrt{26}}\underline{j} - \frac{5}{\sqrt{26}}\underline{k}$

ii. $-\underline{i} + \underline{j} + \underline{k}$

Sol: $\underline{v} = -\underline{i} + \underline{j} + \underline{k}$

$$\left|\underline{v}\right| = \sqrt{(-1)^2 + (1)^2 + (1)^2} = \sqrt{1 + 1 + 1} = \sqrt{3}$$

$$\hat{v} = \frac{\underline{v}}{\left|\underline{v}\right|} = \frac{1}{\sqrt{3}}\left(-\underline{i} + \underline{j} + \underline{k}\right)$$

Given magnitude is7

Required Vector $\underline{a} = 7.\hat{v} = \frac{7}{\sqrt{3}}\left(-\underline{i} + \underline{j} + \underline{k}\right) = \frac{-7}{\sqrt{3}}\underline{i} + \frac{7}{\sqrt{3}}\underline{j} + \frac{7}{\sqrt{3}}\underline{k}$

Q.7. If $\underline{u} = x\underline{i} + 2\underline{j} + 3\underline{k}$, $\underline{v} = \underline{i} + y\underline{j} - 3\underline{k}$ and $\underline{w} = -2\underline{i} - 3\underline{j}$ represent the sides of a triangle Find the values of x and y

Sol: Since $\underline{u}, \underline{v}, \underline{w}$ are sides of Triangle

Therefore $\underline{u} + \underline{v} = \underline{w}$

$$x\underline{i} + 2\underline{j} + 3\underline{k} + \underline{i} + y\underline{j} - 3\underline{k} = -2\underline{i} - 3\underline{j}$$

$$(x+1)\underline{i} + (2+y)\underline{j} = -2\underline{i} - 3\underline{j}$$

$x + 1 = -2$ and $2 + y = -3$

$\Rightarrow x = -3$ and $y = -5$

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Q.8. The position vectors of the points A, B, C and D are $\underline{u} = \underline{i} + 2\underline{j} + \underline{k}$, $\underline{v} = 7\underline{i} + 8\underline{j} + 4\underline{k}$, $\underline{w} = -\underline{i} + \underline{k}$ and $\underline{z} = \underline{i} + 2\underline{j} + 2\underline{k}$ respectively. Show that $\overrightarrow{AB}$ is parallel $\overrightarrow{CD}$

Sol: $\overrightarrow{AB} = \underline{v} - \underline{u} = 7\underline{i} + 8\underline{j} + 4\underline{k} - \underline{i} - 2\underline{j} - \underline{k} = 6\underline{i} + 6\underline{j} + 3\underline{k}$

$\overrightarrow{CD} = \underline{z} - \underline{w} = \underline{i} + 2\underline{j} + 2\underline{k} + \underline{i} - \underline{k} = 2\underline{i} + 2\underline{j} + \underline{k}$

To check if $\overrightarrow{AB}$ is paralleled to $\overrightarrow{CD}$.

$\overrightarrow{AB} = 6\underline{i} \times 6\underline{j} + 3\underline{k} = 3\left(2\underline{i} + 2\underline{j} + \underline{k}\right)$

$\overrightarrow{AB} = 3(\overrightarrow{CD})$

$\overrightarrow{AB}$ is paralleled to $\overrightarrow{CD}$.

نوٹ:

اگر دو ویکٹر آپس میں برابر ہیں یا درمیان میں کوئی constant ہے

$\underline{u} = \lambda \underline{v}$ or $\underline{u} = \underline{v}$ تو Parallel ہو گا

Q.9. We say that two vectors $\underline{v}$ and $\underline{w}$ in space are parallel if there is a scalar c such that $\underline{v} = c\underline{w}$. The vectors point in the same direction if $c > 0$ and the vectors point in the opposite direction if $c < 0$

a) Find two vectors of length 2 parallel to vector $\underline{v} = 2\underline{i} - 4\underline{j} + 4\underline{k}$

b) Find the constant a so that the vectors $\underline{v} = \underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{w} = a\underline{i} + 9\underline{j} - 12\underline{k}$ are parallel.

c) Find a vector of length 5 in the direction opposite that of $\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}$.

d) Find a and b so that the vectors $3\underline{i} - \underline{j} + 4\underline{k}$ and $a\underline{i} + b\underline{j} - 2\underline{k}$ are parallel.

a) Find two vectors of length 2 parallel to vector $\underline{v} = 2\underline{i} - 4\underline{j} + 4\underline{k}$

Sol: $|\underline{v}| = \sqrt{(2)^2 + (-4)^2 + (4)^2} = \sqrt{4 + 16 + 16} = \sqrt{36} = 6$

$\hat{v} = \frac{1}{6}\left(2\underline{i} - 4\underline{j} + 4\underline{k}\right) = \frac{1}{3}\underline{i} - \frac{2}{3}\underline{j} + \frac{2}{3}\underline{k}$

Required vector $= 2\hat{v} = \frac{2}{3}\underline{i} - \frac{4}{3}\underline{j} + \frac{4}{3}\underline{k}$

Vectors in same direction: $+2\hat{v} = \frac{2}{3}\underline{i} - \frac{4}{3}\underline{j} + \frac{4}{3}\underline{k}$

Vectors in opposite direction $-2\hat{v} = -\frac{2}{3}\underline{i} + \frac{4}{3}\underline{j} - \frac{4}{3}\underline{k}$

b) Find the constant a so that the vectors $\underline{v} = \underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{w} = a\underline{i} + 9\underline{j} - 12\underline{k}$ are parallel.

Sol: $\underline{v} = c\underline{w}$

$\underline{i} - 3\underline{j} + 4\underline{k} = c\left(a\underline{i} + 9\underline{j} - 12\underline{k}\right)$

$\underline{i} - 3\underline{j} + 4\underline{k} = ac\underline{i} + 9c\underline{j} - 12c\underline{k}$

Comparison of $\underline{j}$ component $-3 = 9c \Rightarrow c = -\frac{1}{3}$

By comparing $\underline{i}$ component $ac = 1 \Rightarrow a\left(\frac{-1}{3}\right) = 1 \Rightarrow a = -3$

c) Find a vector of length 5 in the direction opposite that of $\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}$.

Sol: $|\underline{v}| = \sqrt{(1)^2 + (-2)^2 + (3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \Rightarrow \hat{v} = \frac{1}{\sqrt{14}}(i - 2j + 3k)$

Required vector $= -5.\hat{v} = \frac{-5}{\sqrt{14}}\left(\underline{i} - 2\underline{j} + 3\underline{k}\right) = \frac{-5}{\sqrt{14}}\underline{i} + \frac{10}{\sqrt{14}}\underline{j} - \frac{15}{\sqrt{14}}\underline{k}$

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d) Find a and b so that the vectors $3\underline{i} - \underline{j} + 4\underline{k}$ and $a\underline{i} + b\underline{j} - 2\underline{k}$ are parallel.

Sol: $\underline{v} = c.\underline{w}$

$\Rightarrow 3\underline{i} - \underline{j} + 4\underline{k} = c\left(a\underline{i} + b\underline{j} - 2\underline{k}\right)$

$\Rightarrow 3\underline{i} - \underline{j} + 4\underline{k} = ac\underline{i} + bc\underline{j} - 2c\underline{k}$

By comparison of component

$3 = ac$ _____ (I), $-1 = bc$ _____ (II) and $4 = -2c \Rightarrow c = -2$

Put values of c in I and II

$3 = (-2)a$ $\qquad -1 = b(-2)$

$\Rightarrow a = \dfrac{-3}{2}$ $\qquad \Rightarrow b = \dfrac{1}{2}$

Q.10 A spacecraft moves from point $(120, 240, -50)$ to point $(130, 210, 80)$ in kilometers. What is the magnitude of the displacement vector in kilometers?

Sol: Final point $= (130, 210, 80)$

Initial point $= (120, 240, -50)$

Displacement $= (130-120, 210-240, 80-(-50)) = (10, -30, 130)$

$|\underline{d}| = \sqrt{(10)^2 + (-30)^2 + (130)^2} = \sqrt{100 + 900 + 16900} = \sqrt{17900}$

$|\underline{d}| = 10\sqrt{179}\ km$

Q.11 Find the direction cosines for the given vector:

i. $\underline{u} = -6\underline{i} + 3\underline{j} + 2\underline{k}$ ii. $\underline{v} = 4\underline{i} + 2\underline{j} - 5\underline{k}$ iii. $\overrightarrow{PQ}$, where $P(9,3,13)$ and $Q(11,6,19)$

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i. $\underline{u} = -6\underline{i} + 3\underline{j} + 2\underline{k}$

Sol: $|\underline{u}| = \sqrt{(-6)^2 + (3)^2 + (2)^2} = \sqrt{36 + 9 + 4} = \sqrt{49} = 7$

Direction cosines are $\cos\alpha = -\dfrac{6}{7}$, $\cos\beta = \dfrac{3}{7}$, $\cos\gamma = \dfrac{2}{7}$

ii. $\underline{v} = 4\underline{i} + 2\underline{j} - 5\underline{k}$

Sol: $|\underline{v}| = \sqrt{(4)^2 + (2)^2 + (-5)^2} = \sqrt{16 + 4 + 25} = \sqrt{45} = 3\sqrt{5}$

Direction cosines are

$\cos\alpha = \dfrac{4}{3\sqrt{5}}$, $\cos\beta = \dfrac{2}{3\sqrt{5}}$, $\cos\gamma = \dfrac{-5}{3\sqrt{5}} = -\dfrac{\sqrt{5}}{3}$

iii. $\overrightarrow{PQ}$, where $P(9,3,13)$ and $Q(11,6,19)$.

Sol: $\overrightarrow{PQ} = (11,6,19) - (9,3,13) = (11-9, 6-3, 19-13) = (2,3,6)$

$\overrightarrow{PQ} = 2\underline{i} + 3\underline{j} + 6\underline{k}$

$\left|\overrightarrow{PQ}\right| = \sqrt{(2)^2 + (3)^2 + (6)^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$

Direction cosines are $\cos\alpha = \dfrac{2}{7}$, $\cos\beta = \dfrac{3}{7}$, $\cos\gamma = \dfrac{6}{7}$

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Q.12. Which of the following triple can be the direction angles of a single vector?

i. $45°, 45°, 60°$ ii. $30°, 45°, 60°$ iii. $45°, 60°, 60°$

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i. $45°, 45°, 60°$

Sol: Here $\cos\alpha = \cos 45° = \frac{1}{\sqrt{2}}, \cos\beta = \cos 45° = \frac{1}{\sqrt{2}}, \cos\gamma = \cos 60° = \frac{1}{2}$

Now $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$$= \cos^2 45° + \cos^2 45° + \cos^2 60° = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} = 1 + \frac{1}{4} = \frac{5}{4} \neq 1$$

These angles cannot be the direction angles of a single vector.

ii. $30°, 45°, 60°$

Sol: Here $\cos\alpha = \cos 30° = \frac{\sqrt{3}}{2} \Rightarrow \cos\beta = \cos 45° = \frac{1}{\sqrt{2}} \Rightarrow \cos\gamma = \cos 60° = \frac{1}{2}$

Now $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$$= \cos^2 30° + \cos^2 45° + \cos^2 60° = \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{3}{4} + \frac{1}{2} + \frac{1}{4} = \frac{3}{4} + \frac{2}{4} + \frac{1}{4} = \frac{3+2+1}{4} = \frac{6}{4} = \frac{3}{2} \neq 1$$

These angles cannot be the direction angles of a single vector.

iii. $45°, 60°, 60°$

Sol: Here $\cos\alpha = \cos 45° = \frac{1}{\sqrt{2}} \Rightarrow \cos\beta = \cos 60° = \frac{1}{2} \Rightarrow \cos\gamma = \cos 60° = \frac{1}{2}$

Now $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

$$= \cos^2 45° + \cos^2 60° + \cos^2 60° = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = \frac{2}{4} + \frac{1}{4} + \frac{1}{4} = \frac{2+1+1}{4} = \frac{4}{4} = 1$$

Since the sum of the squares of the cosines is 1, these angles are the direction angles of a single vector.

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